EVAN ZHANG

Portfolio: evanrzhang.github.io

San Jose, CA evan.r.zhang@gmail.com 408-333-1816

EDUCATION

University of Pennsylvania, *Jerome Fisher Program in Management and Technology* **August 2026 - May 2030**

- *School of Engineering and Applied Science:* BSE in Electrical Engineering, MSE in Mechanical Engineering
- *The Wharton School:* BS in Economics, Concentration in Finance

The Harker School, San Jose, CA **August 2022 - May 2026**

- **ACT:** 36, **Cumulative GPA:** 4.35/4.0
- **Advanced Topics Courses:** Honors Linear Algebra, Honors Multivariable Calculus, AP Physics C Mechanics, AP Physics C Electricity and Magnetism, Honors Advanced Research, Honors Nanoscience, AP Calc BC

SKILLS

- **Design:** SolidWorks, Onshape, KiCad (PCB Design); **Fabrication/Machining:** 3D printing, CNC; **FEM/CFD:** Abaqus, Ansys, XFLR5; **Programming:** Java, Python (Machine Learning), C, Mathematica, LaTeX, LabVIEW, Computer Vision (OpenCV), MATLAB; **Robotics & Control Systems:** MyRIO, Arduino, Robot Operating System (ROS)

EXPERIENCE

***Researcher & Start-up Founder & CEO*: Enviro-AI, developing "SPIRo"** **June 2023 - Present**

- Designed a pneumatic, modular soft robot, SPIRo, with a machine learning algorithm for inspecting gas leaks.
- Founded VC-backed start-up Enviro-AI, secured the 1517 Fund Medici Grant, Nonprovisional Patent (published in Jan 2026). 2 first-author publications, 2 second-author publications

***Research Intern*, Chortos Lab at Purdue University** **June 2024 - June 2026**

- Developed ML-controlled, 3D-printed HASEL soft actuators for cardiac assist devices. Integrated sensing, ML, and FEM/CFD modeling to optimize control and response.

***Summer Research Intern*, The Faboratory, Yale University** **July - August 2025**

- Built a shape-morphing amphibious robot with variable stiffness limbs, enabling adaptive locomotion.
- Proposed a deep-water capable robotic turtle, designed high-pressure waterproof casings for motors.

***Mechanical Engineering Intern*, Schnitzer Lab at Stanford University** **Dec 2022 - June 2024**

- Designed circuit schematics and PCBs for sensing and control of experimental apparatus used across the lab.
- Implemented high-throughput NAS storage infrastructure to handle large volumes of imaging data.

SELECTED PATENTS, PUBLICATIONS, AND AWARDS

- **Regeneron Science Talent Search** (STS) Top 300 Scholar (January 2026)
- 2-Time **International Science and Engineering Fair (ISEF) Finalist** & Grand Prize 4th Place (2025, 2026)
- Conrad Challenge, **Pete Conrad Scholar** and 1st place in Energy and Environment category (April 2025)
- **Coca-Cola Scholars** Semi-Finalist (2025)
- 3D-Printed HASEL Actuators With Integrated Fluidic Piezoresistive Pressure Sensor for Cardiac Assist Applications (ICMSR 2025, December 2025) **Evan Zhang**, Alex Chortos
- Omnimobile and Load Capable Pneumatic Soft Robot for Gas Distribution Pipeline Inspection (IEEE Robosoft 2025, April 2025, Full Oral Talk) **Evan Zhang**, Eddie Zhang
- Gas Pipeline Leakage Detection Based on Multiple Multimodal Deep Feature Selections and Optimized Deep Forest Classifier (Frontiers in Environmental Science, April 2025) Eddie Zhang, **Evan Zhang**
- **Patent #19/274609**: SPIRo: An AI-Based Pneumatic Soft Robot with Machine Learning-Driven Multidimensional Locomotion and Multimodal Deep Feature Selection and Fusion with an Improved Deep Forest Classifier Architecture for Gas Pipeline Leak Detection (January 2026) **Evan Zhang**, Eddie Zhang

OTHER LEADERSHIP EXPERIENCES

***Founder and President*, Harker 3D Printing and Design Club** **June 2024 - June 2026**

- Founded and scaled the club to 60+ members, taught CAD, and trained members on the 3D printer.

***Co-President,* Harker Aerospace Club** **Aug 2022 - June 2026**

- Founded and led 2 teams to compete in the American Rocketry Challenge, grew the club to 70+ members.

***Mechanical & Electrical Director,* FTC 11311/ *Design Team*, Harker FRC 1072** **Aug 2021 - August 2024**

***Captain and Coach*, Harker Varsity Debate Team** **Aug 2022 - June 2026**